

Automated Classification of Steel Surface Defects Using Convolutional Neural Networks

1. Motivation and Background

In high-volume manufacturing such as aerospace or steel production, surface quality inspection is critical. Traditional inspection is manual, inconsistent, and error-prone. Deep learning—especially CNNs—provides a scalable, accurate alternative for visual defect classification. The NEU Surface Defect Database provides a standardized dataset for studying defect classification.

2. Problem Statement

Goal: Given a grayscale image of a steel surface, classify it into one of six defect categories. Research questions include assessing model performance, the effects of regularization, and sources of misclassification.

3. Dataset Description

Dataset: NEU Surface Defect Database (public Kaggle mirror). Contains 1,800 grayscale 200×200 images labeled into six classes: crazing, inclusion, patches, pitted surface, rolled-in scale, scratches.

4. Methodology

Approach: Data split (70/15/15), normalization, and augmentation. Baselines include logistic regression and a simple MLP. The primary CNN uses three convolutional blocks followed by a dense layer and softmax output. Refinements will include batch normalization, dropout, and learning-rate scheduling. Transfer learning (e.g., ResNet50) may be explored.

5. Training and Evaluation

Loss: categorical cross-entropy; Optimizer: Adam; Epochs: ~50; Metrics: accuracy, precision, recall, F1-score. Validation curves and confusion matrices will support interpretation and refinement.

6. Alignment with COMP 4531 Learning Objectives

Demonstrates understanding of neural network design, CNN architecture, optimization, and TensorFlow/Keras implementation per course objectives. Fits under the 'Image Recognition' rubric category.

7. Tools and Resources

Language: Python 3.x; Framework: TensorFlow/Keras; Hardware: local macOS Metal GPU or Google Colab. Dataset size is manageable for laptop training.

8. Risks and Limitations

Dataset size may limit generalization. Project focuses on classification (not localization). All data used are public and unclassified.

9. Expected Outcomes

Deliverables: (1) Written report detailing methodology and results, (2) Jupyter notebook with code and visualizations, (3) Presentation summarizing process, performance, and findings.